Transferring Heat (Hopefully)

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ENGR 360: Heat Transfer Liberty University School of Engineering

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23 November 2024

An optimal shell and tube heat exchanger was designed by the team to meet specifications from the customer and was analyzed for heat transfer, efficiency, and pumping power. This design included a 9-meter long counterflow heat exchanger with 38 tubes and 0.05m diameter. By making simplifying assumptions to iterate a clear solution during calculations, the team was able to assert this conclusion. The effectiveness NTU method was used during calculations. Through this method, the optimized design worked to maximize the amount of heat transfer between the two fluids, while also minimizing area and minimizing power. The design was then analyzed in ANSYS Fluent with a simplified one tube heat exchanger design to compare results found in the geometric calculations.

Keywords—Heat exchanger, ANSYS, Effectiveness NTU

I. Introduction

The team was asked to design and analyze a heat exchanger that would maximize heat transfer between the hot and cold fluid in the exchanger, while also minimizing area and power consumed. While keeping this in mind, there were also constraints given by the customer that needed to be considered. These constraints included a mass flow rate for the hot and cold water, an inlet temperature for both, and an outlet temperature for the cold water. These specific considerations are shown in Table I below. In addition, a length constraint was also given to ensure the total length is between 1 and 10 meters. The one shell and one tube pass heat exchanger design that was optimized and analyzed further through calculations and ANSYS simulation is a counterflow heat exchanger that includes 38 tubes within the shell. The hot water would flow within the tubes and the cold water would flow through the gap in the shell to maximize the amount of heat exchange. Fins and baffles were considered for the design as they increase heat rate and efficiency [1] but was decided against pursuing further because area would not be minimized using this method. The calculations used during analysis utilized the effectiveness NTU method which is generally used when outlet temperature are not given. This method goes through the process of finding the minimum heat capacity rate, calculating the NTU using the overall heat transfer coefficient, the area of the pipe, and the capacity rate found to then ultimately find max heat rate. This then provides the temperatures that were not given. All equations and tables used in the geometric analysis and heat exchanger analysis are from The Fundamentals of Heat and Mass Transfer unless otherwise started and are in Appendix A [2]. Assumptions used during the geometric calculations included assuming steady state, incompressible fluid, fully developed flow, constant fluid properties, constant pressure specific heat, turbulent flow, constant mass flow rate, insulated shell, same pressure drop across all pipes, radiation is neglected, overall heat transfer coefficient is constant, friction factor is constant, no changes in kinetic or potential energy, and the friction of the shell is negligible. Following the geometric calculations and heat exchanger analysis, ASYS simulations were conducted on the 9-meter-long heat exchanger, where values of effectiveness, maximum heat rate, NTU, and temperature of the hot water outlet for both the hand calculations and ANSYS results were then compared. The outlet temperature, effectiveness, and NTU results were very similar in both the hand calculations but varied in terms of maximum heat rate. The reasoning behind this will be discussed further.

TABLE I

	$\dot{m}\ (\frac{kg}{s})$	$T_{m,i}$ (°C)	$T_{m,o}$ (°C)
Hot Water	28	90	-
Cold Water	27	34	60

Constraints and values given by customer

II. Geometric Calculations

A. *Pipe Dimensions*

When producing the geometric dimensions of the heat exchanger, iteration was needed. Initial values for the number of tubes N, the diameter of inner pipes D_i, and the length L were chosen. The following required variables, including cross-sectional area A_c, surface area A_s, pressure loss per pipe Δp, and total pump power $\dot{W}$, were solved for in MATLAB, as shown in Appendix B. The geometric values were altered one by one until reaching optimization for the required variables. Optimizing this design included maximizing the heat transfer coefficient U and minimizing the area of total heat exchanged. For a summary of the pipe dimensions found, see Table II.

When evaluating copper types, L copper pipes were chosen because they are considered intermediate in thickness, weight, and size range when compared to M and K copper pipes [3][4]. Different sizes are seen in appendix D photo 1. Using interpolation between pure and commercial bronze copper, the properties of density at ρ_{cu} = 8900 and thermal conductivity at k_{cu} = 398 are given by Table A.1.

For readability and so that required values may be presented below, the ultimate geometric variables are displayed here. The final iteration provides the number of

tubes of N = 38, the length of L = 9 m, inner tubes with the diameter of D_i = 0.05 m and the outer diameter, to allow for copper thickness, of D_{i2} = 0.05398 m. These diameters are typical L pipe sizes (see appendix D photo 1).

Within the for loop, the following variables are present, with the final results shown below. To allow space for the inner tubes, the shell diameter of the pipe needed to be 1.2 m. To account for this when iterating, the diameter of the inner tubes were by multiplied by a scaling coefficient of 24. The cross-sectional area of the small tubes and of the shell were solved by multiplying pi by the squared radius of the inner tubes or shell, respectively. This gave the inner tubes a cross-sectional area of $A_{c,i} = 1.96(10^{-3})$ m^2 and the outer pipe's shell a cross-sectional area of $A_{c,o} = 1.13$ m^2. The surface areas of the inner tubes and the shell were calcualted from the product of the number of tubes, pi, length, and the inner or shell diameter. The surface area of the inner pipes is $A_{s,i} = 57.56$ m^2 and the surface area of the outer pipe is $A_{s,o} = 33.93$ m^2. It is important to note, because of the initial assumption that the shell is insulated, allowing no heat transfer, $A_{s,o}$ will not be used in further calculations, such as when finding the total thermal resistance.

TABLE II

Dimension	**Value**	**Units**
Number of tubes, N	38	—
Length, L	9	m
Shell Diameter, D_o	1.20	m
Inner Diameter of inner tubes, D_i	0.0500	m
Outer Diameter of inner tubes, D_{i2}	0.05398	m
Cross-Sectional Area of Inner Pipes, $A_{c,i}$	$1.96(10^{-3})$	m^2
Cross-Sectional Area of Outer Pipe, $A_{c,o}$	1.13	m^2
Surface Area of Inner Pipes, $A_{s,i}$	58.00	m^2
Surface Area of Outer Pipe, $A_{s,o}$	33.93	m^2

Geometric calculation values.

B. *CAD Drawing*

To visualize the designed heat exchanger from above, a SolidWorks model was made. This model fits all the geometric specifications calculated in Section IIA. This includes 38 inner tubes, a length of 9 m, a shell diameter of 1.2 m, an outer diameter of the small tubing of D_{i2} = 0.05398 m and an inner dimater of the small tubing of 0.0500 m.

The SolidWorks drawing showing these dimensions is shown in Figure 1. Additionally, two renderings are provided in Figure 2 and Figure 3.

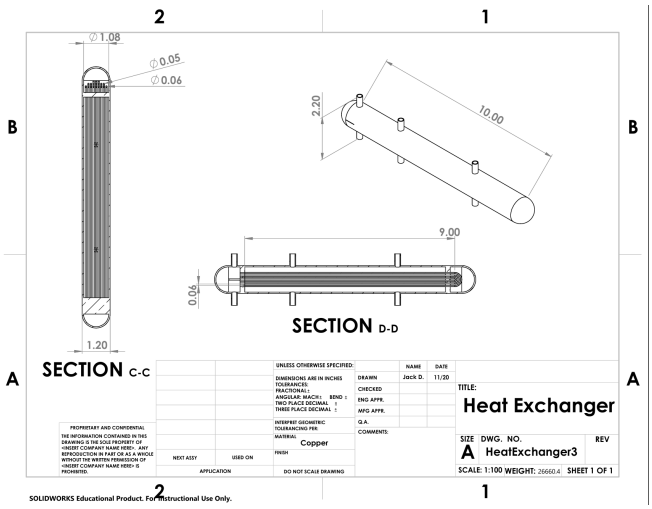


Fig. 1. SolidWorks drawing of heat exchanger. To look at dimensions, see enlarged view in Appendix C.

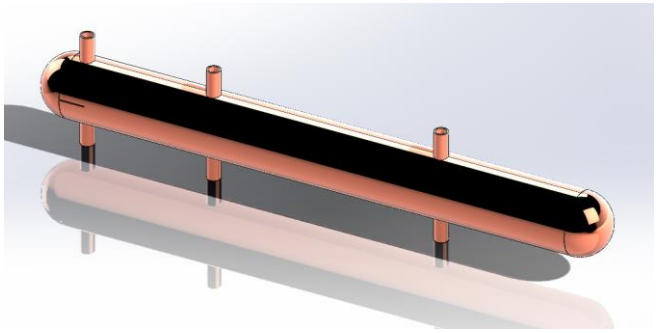
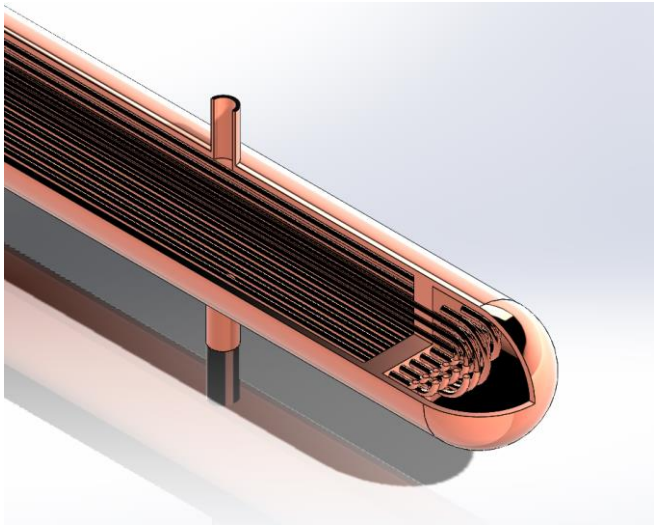


Fig. 2. SolidWorks rendering of the entire heat exchanger.



*F*ig. 3. Partitioned rendering of SolidWorks model to observe inside tubing.

III. HEAT EXCHANGER ANALYSIS

A. *Outlet Hot Temperature and Heat Rate*

For a more concise display of the values calculated in the heat exchanger analysis, Table III is provided.

The effectiveness-NTU method was used when solving for the requested heat exchanger properties. This method required multiple variables and properties excluding what was explicitly asked for by or given from the client. Using the average cold-water temperature of $T_{C,avg}$ = 320 K, a specific heat of $c_{p,c}$ = 4180 J/kg·k was found. The given cold-water outlet temperature of $T_{c,o}$ = 333 K provided a kinematic viscosity of $\mu_c = 0.000467\ m^2/s$, a thermal conductivity of $k_c = 0.654$ W/mK, and a Prandtl number of $Pr_c = 2.99\ m^2/s$ with Table A.6. With the values given and the values found above, eq. 11.7b, shown in Appendix B, was utilized to find the heat transfer rate of the cold side, q_c = 2934.46 kW.

In order to find the specific heat for the hot side $c_{p,h}$, an initial value for hot outlet temperature $T_{h,o}$ was chosen to be 350 K. With this starting value and the given hot inlet temperature supplying a mean temperature, Table A.6 data was interpolated to find $c_{p,h}$, which then in turn was used to find a more accurate $T_{h,o}$ using eq. 11.6. This cycle was iterated using MATLAB for a maximum of one thousand times or until the temperature difference between iterations was minimal, meaning there was less than a 0.00001 K difference. The actual $T_{h,o}$ was discovered to be 338.02 K. Table A.6 at this hot-water outlet temperature provides a kinematic viscosity of $\mu_c = 4.33(10^{-4})$, a thermal conductivity of k_c = 0.658, and a Prandtl number of Pr_c = 2.75.

By referring to eq. 11.26, shown below, the heat capacity rates of the hot and cold sides were evaluated using the assigned mass flow rates. Comparing the two showed that C_{min} is $C_c = 1.129(10^5)$, C_{max} is $C_h = 1.175(10^5)$, and the heat capacity ratio C_r = 0.9607 (page 510). Equation 11.18 gives q_{max} = 6.32 MW by multiplying the minimum heat capacity rate by the difference in inlet temperatures.

$$\frac{C_{\min}}{C_{\max}} = \frac{\dot{m}_h c_{p,h}}{\dot{m}_c c_{p,c}} = \frac{T_{c,o} - T_{c,i}}{T_{h,i} - T_{h,o}} \qquad (11.26)$$

B. *Effectiveness and NTU Calculations*

Effectiveness was found using the q_{max} with eq. 11.19, giving ε = 0.464. In order to determine whether parallel or counter flow performs better, the UA ideal values are found and compared. Table 11.42 provides Eq. 11.28b and Eq 11.29b for the NTU values in both flow arrangements using C_r and ε, with the parallel flow value being NTU_{pf} = 1.23 and the counter flow value being NTU_{cf} = 0.852. The ideal UA for parallel and counter flow are found, from eq. 11.24, to be $1.388(10^5)$ and $9.618(10^4)$, respectively. Using the same equation, the NTU value for both cases was multiplied by the C_{min} and divided by U_{min} of 850 or U_{max} of 1700, given in Table 11.2. Following this, the areas for the parallel flow were $A_{pf,min}$ = 81.66 m^2 and $A_{cf,max}$ = 163.32 m^2, and the areas for the counter flow were $A_{cf,min}$ = 56.58 m^2 and $A_{cf,max}$ = 113.16 m^2. Because counter flow has a smaller required cross area to get the same heat transfer rate compared to parallel flow, the counter flow heat exchanger method was chosen.

$$\mathrm{NTU} = -\frac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r} \qquad (11.28b)$$

$$\mathrm{NTU} = \frac{1}{C_r - 1}\ln\left(\frac{\varepsilon - 1}{\varepsilon C_r - 1}\right) \quad (C_r < 1) \qquad (11.29b)$$

C. *Pressure and Power Loss Calculations*

Finally, the pressure and power loss were calculated. Because these values change with the geometric parameters, they are found within the for loop in MATLAB mentioned in Section II.A.

Starting with Eq. 8.5, the velocity of an inner pipe was found using the given $\dot{m}_h$ divided by the number of tubes, giving V_i = 0.383 m/s. This value was used later on when finding the pressure drop. The Reynolds numbers were found through the tubes and the shell using Eq. 8.6 and the fluid properies found previously, providing values of Re_i = $1.65(10^6)$ and $Re_o = 6.13(10^4)$, respectively. These values support the initial assumption made that the flow throughout the heat exchanger was considered turbulent. In order to find the convection coefficients, the Nusselt numbers for both the tubes and the shell were found using the Dittus-Boelter equation (Eq. 8.60), with Nu_i = 2928.4 and Nu_o = 240.9. For the shell convection coefficient, the hydraulic diameter must be found, using Eq. 8.66, as the corrected outer tube diameter. This accounted for the imaginary diameter's cross-sectional area, giving D_h = 0.05668 m. Equation 8.53 provided values of $h_i = 3.86(10^4)$ W/m^2·k and $h_o = 2.78(10^3)$ W/m^2·k for the inner and shell convection coefficients.

The total resistance R_{tot} is known to be equivalent to the inverse product of the overall heat transfer coefficient U and the surface area A_s and is found through a summation of the different thermal resistances, as shown in Eq. 11.5 below. These different resistances are calculated using the convection coefficients for the inner tubes and the shell. After finding that $R_{tot} = 1.45(10^{-5})$ K/W, the overall heat transfer coefficient U was found by taking the product of R_{tot} and the surface area of the inner tube, with $U = 1.19(10^3)$ W/m^2·k. Referring to Section III.B, the ideal *UA* was divided by the *U* found above to find the total surface area of the heat exchanger, A_s = 80.67 m^2. It is important to note that this surface area fulfills the counter flow minimum and maximum surface area constraints that were calculated in Section III.B.

$$\frac{1}{UA} = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} = \frac{1}{h_i A_i} + \frac{R''_{f,i}}{A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{R''_{f,o}}{A_o} + \frac{1}{h_o A_o} \qquad (11.5)$$

After research, the surface roughness of copper was found to be ε = 0.0003 m [5]. This value divided by the inner tube diameter gives the dimensionless roughness ε/D = $6.00(10^{-4})$ which can be used concurrently with the Reynolds number Re_i on the Moody diagram to find the friction factor *f*. Equation 8.20 can also be used to find this value. Using both methods for accuracy, *f* = 0.0177.

To find the pressure drop per pipe, Eq. 8.22b is used and gave Δp = 228.4 kPa, or when multiplied by the number of tubes, gave a total pressure drop of Δp_{tot} = 8.68 kPa. Taking the total pressure drop, Eq. 8.22b, which multiplies it by the volumetric flow rate, gave a total pump power of P_{tot} = 239.6 W. This is equivalent to 6.31 W per pipe.

TABLE III

Dimension	**Value**	**Units**
Outlet Hot Temperature, $T_{h.o}$	338.02	K
Heat Rate, q	6.32	MW
Effectiveness, ε	0.464	—
NTU_{cf}	0.852	—
Pressure Loss per pipe, Δp	228.43	Pa
Pumping Power, $\dot{W}$	239.6	W

Heat exchanger analysis values.

IV. ANSYS FLUENT SIMULATION

Upon instruction from the customer's requirements, an ANSYS Fluent simulation was conducted. For this simulation, a simplified version of a heat exchanger was modeled, incorporating a single pipe in an annulus. This pipe located in the middle of the annulus was modeled to contain the hot fluid, while the surrounding area, the outer layer of the annulus contains the cold fluid during the simulation. From the project instructions, during the creation of this simulation, the generation of contour plots regarding static pressure, temperature, and mass flow rate were created. These variables would be analyzed using a small slice of the model to gather information about the heat exchanger's capabilities regarding factors such as pressure, temperature, and mass flow rate. This includes how these three variables changed over time. A plot of temperature vs. position was requested which shows the change of temperature according to position at the center of the hot fluid, and at the center of the space where the cold fluid is contained. In other words, the center of the space where the cold fluid is located would be half of the at the outer diameter minus the inner diameter. A plot representing these changes was graphed with position being the independent variable, while temperature was dependent on the position. In addition to this, iterations of the simplified heat exchanger mesh were conducted with varying types of mesh, inlet and outlet layers, total thickness, mesh quality, and many more factors.

The geometry of the heat exchanger was made according to the constraints listed above, incorporating the hot liquid pipe in the center of the annulus, with the cold liquid surrounding this area on the outside with a specified outer diameter. However, this design was simplified from our design created using mathematical analysis. Following the results of calculations, the optimized heat exchanger came out to be 9 meters, however the initial ANSYS simulation was conducted on a 0.5-meter-long geometry instead of the full 9 meters. After further clarification, the information used during this initial analysis was then used to create a simulation on the full 9-meter-long heat exchanger, which shows the final product. The geometry was modeled beginning with the shell, then inserting the tube within the existing shell in the center, according to a specified inner diameter. This tube's purpose was to contain the hot fluid. After these components were modeled, the inner and outer flows were added using the fill feature.

After the geometry was made for the exchanger, the mesh iterations were conducted using the 0.5-meter simplified geometry. The purpose of this mesh was to separate the geometry into smaller elements or nodes that allows the simulation of flow to conclude more accurate results. ASNYS analyzes the flow through each of these nodes, which all have a specific shape according to the geometry and completes calculations using equations associated with each node. Refinements were added to the mesh during this process in the addition of inflations. Inflations were added to the inner and outer flows of the exchanger, and boundaries were added to the outer area of the geometry. Initially, a total thickness inflation was added, beginning with 15 layers, a growth rate of 1.2 and a maximum thickness of 0.004 meters. These characteristics result in the use of layers among the geometry all along the tube of the exchanger. Thirteen iterations following this process were then conducted with varying mesh types, altering layer amounts, growth rates, and maximum thicknesses. For all these mesh iterations, inflation was the method of mesh refinement chosen. Six of these iterations consisted of using a total thickness inflation method, six of them consisted of a smooth transition inflation method, and one included a control auto mesh. Following these iterations, the mesh quality of each was determined in terms of average orthogonal quality, max skewness, and average skewness. All iterations conducted had an average orthogonal quality between 0.729 and 0.775, which are optimal results because these are higher than 0.2. Also, the max skewness for all iterations resulted in values between the range of 0.832 to 0.899. After initial values were determined, further analysis was conducted to figure out convergence, and temperature difference between inlet and exit on both the pipe with hot water and shell with cold water. This was important to find the heat transfer that was done throughout the exchanger. Through this process, the boundary conditions as well as the material of the pipe, which was copper, were defined. Six iterations were conducted using the given constraints of mass flow rate, and inlet temperatures, and all six tests converged. In addition, the temperature differences were determined, and test number three of this set resulted in the highest difference for the cold water. Therefore, this showed the highest amount of heat transfer out of all of the iterations conducted. The results of these seven tests are shown below in Tables IV and V.

TABLE IV

Mesh #	**Type**	**Cold + Hot Pipe Layers**	**Thickness**

1	Inflation, Total Thickness	5	0.002
2	Inflation, Total Thickness	10	0.002
3	Inflation, Total Thickness	15	0.004
4	Inflation, Smooth Transition	5	_
5	Inflation, Smooth Transition	10	_
6	Inflation, Smooth Transition	15	_

Initial mesh iteration data

TABLE V

Mesh #	Orthogonal Quality	Hot Temperature Difference (°C)	Cold Temperature Difference (°C)
1	0.75046	8.946285	8.688882
2	0.77524	8.880945	8.688341
3	0.74581	8.930169	8.684918
4	0.74377	8.858923	8.712488
5	0.75592	8.752236	8.887473
6	0.74716	8.759718	8.935048

Initial mesh iteration data continued

As stated above, once simulations were conducted on the shorter, 0.5-meter heat exchanger, full simulations were conducted on the optimized 9-meter heat exchanger that was found to be best during geometric calculations. The iterations completed above were used to find the best mesh type, thickness, growth rate, and thickness level which was then used for the full simulation and analysis. Once the updated geometry was made, which is shown below in Fig. 4, contour plots of static pressure, temperature, and mass flow rate were then generated. These plots are shown below in Fig. 5-7, as well as the requested plot of position vs. temperature in Fig. 8.

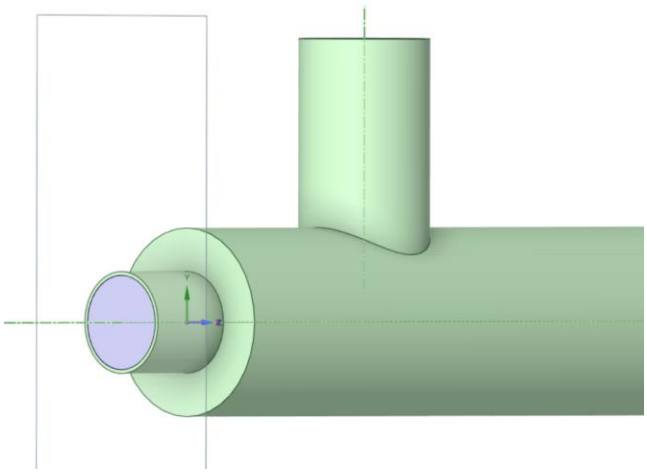


Fig. 4. Partial Image of Geometry

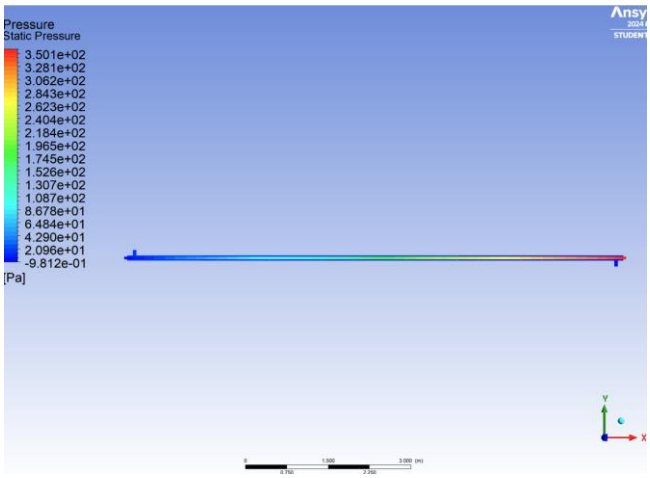


Fig. 5 Static Pressure Contour Plot

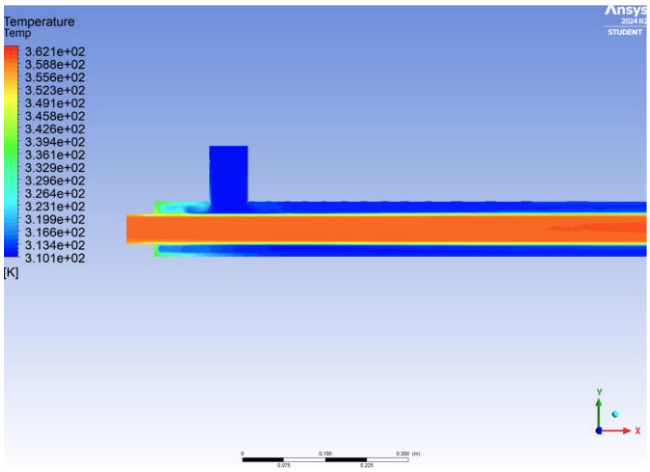


Fig. 6. Temperature Contour Plot

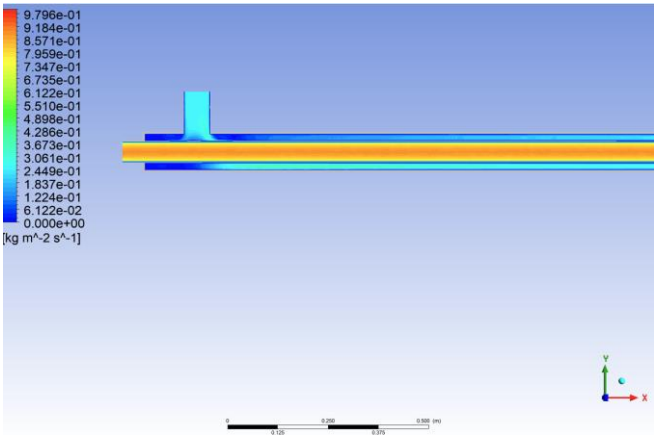


Fig. 7. Mass Flow Rate Contour Plot

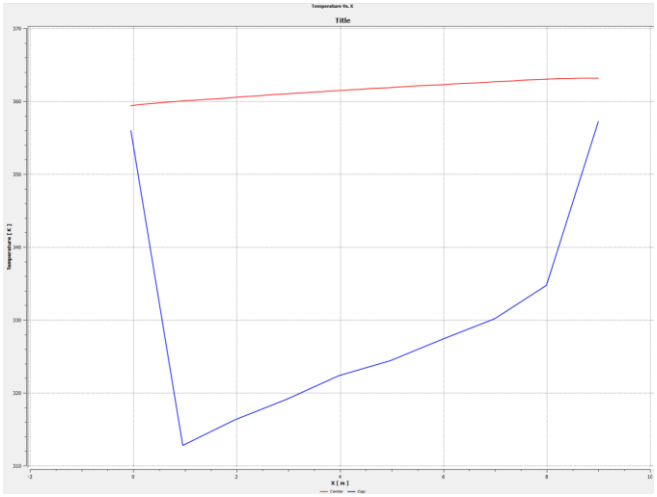


Fig. 8. Position vs. Temperature at center of hot fluid and center of the gap of cold fluid

Using the information gathered from the simplified heat exchanger simulations, mesh iterations were made following the same steps. A total of five iterations were made with this full-length geometry, including one control mesh and four with inflation refinements with a smooth transition. The layers of these varied between 2 and 15 layers, and the growth rate remained constant at 1.2 for both the hot pipe and cold shell for all tests. The nodes varied among these, ranging from 237,440 nodes to as high as 1,279,833 nodes. Due to licensing issues the iterations that included layers over two did not converge, however the iteration with two layers for both the cold shell and hot pipe ended up converging. Full results of the four iterations that included inflation, and a smooth transition are included below in Table VI.

TABLE VI

Mesh #	Cold Pipe	Hot Pipe	Nodes	Converge?
1	15 Layers	15 Layers	237440	No
2	5 Layers	5 Layers	2353902	No
3	3 Layers	3 Layers	1309876	No
4	2 Layers	2 Layers	1019518	Yes

Mesh iteration data for full heat exchanger

Following the mesh iterations made on the full-length exchanger, code that is shown in Figure 9 was used to compare similarities and differences between the hand calculated values found during the geometric calculations and the ANSYS Fluent simulations. These values include maximum heat rate, NTU, mass flow rate, and various others. The outputs of these values are discussed later in further detail.

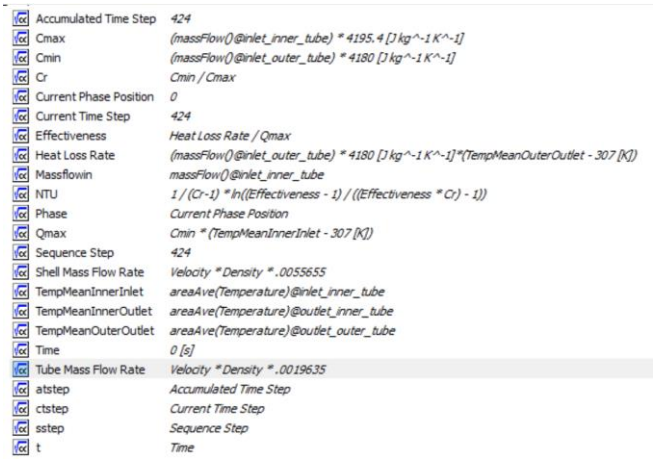


Fig. 9. Formulas used in ANSYS to find requested variables

V. DISCUSSION

When analyzing the results, a length of 9 meters, 38 inner tubes, and a diameter of D_i = 0.05 m provides an effective heat transfer rate, of q_{max} = 6.32 MW, while meeting the restraints given by the client. In addition to ANSYS Fluent supporting the data, the values found with effectiveness-NTU were satisfied in internal flow calculations to ensure consistent effectiveness, such as UA. The geometry iterations prioritized maximizing the heat transfer coefficient *U* while minimizing the area of total heat exchanged A_s.

After ANSYS Fluent simulations were completed, variables were solved such as the temperature of the hot water outlet, maximum heat rate, effectiveness, and NTU. These values were compared to the hand calculations done during the heat exchanger analysis. Variables like hot water outlet temperature, effectiveness, and NTU were relatively similar in both cases, however, the maximum heat rate varied significantly. See appendix D photo 2, where the purple results are from ANSYS, and blue are from MATLAB. This is due to the ANSYS simulation only having 1 pipe while the actual heat exchanger has 38 inner pipes. The specific differences are shown in Table VII below. During the ANSYS iterations conducted, limitations were encountered due to the high cell counts. With the current ANSYS license,

cell counts to 1,100,000 will converge, however due to this restriction, iterations with layers more than 2 could not be done. Therefore, the most useful iteration completed included 2 iterations, and converged.

There are ethical considerations needed when designing the heat exchanger. The first is ensuring efficiency to reduce energy consumption while lowering the carbon footprint. The next is the environmental impact. The material needs to be eco-friendly to prevent damage to the surrounding environment. And finally, safety and reliability, which is done by ensuring correct lengths and consistent numbers through calculations.

TABLE VII

	Hand Calculations	**ANSYS**
Temperature of Hot Water Outlet (K)	338.02	358.73
Effectiveness	0.4643	0.5845
NTU	0.852	0.941
Maximum Heat Rate (W/m^2)	6320160	33783.7

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APPENDIX A

Table A.1: Equations used in heat exchanger calculations

$R_{t,\text{cond}} = \frac{\ln\,(r_2/r_1)}{2\pi L k}$	3.33	$q = \dot{m}_c c_{p,c}(T_{c,o} - T_{c,i})$	11.7b
$\dot{m} = \rho u_m A_c$	8.5	$q_{\max} = C_{\min}(T_{h,i} - T_{c,i})$	11.18
$Re_D = \frac{4\dot{m}}{\pi D \mu}$	8.6	$\varepsilon \equiv \frac{q}{q_{\max}}$	11.19
$Nu_D \equiv \frac{hD}{k}$	8.53	$\text{NTU} \equiv \frac{UA}{C_{\min}}$	11.24
$Nu_D = 0.023\, Re_D^{4/5} Pr^n$	8.60	$\frac{C_{\min}}{C_{\max}} = \frac{\dot{m}_h c_{p,h}}{\dot{m}_c c_{p,c}} = \frac{T_{c,o} - T_{c,i}}{T_{h,i} - T_{h,o}}$	11.26
$D_h \equiv \frac{4A_c}{P}$	8.66	$\text{NTU} = -\frac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$	11.28b
$\frac{1}{UA} = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} = \frac{1}{h_i A_i} + \frac{R''_{f,i}}{A_i} + \frac{\ln(D_o/D_i)}{2\pi k L} + \frac{R''_{f,o}}{A_o} + \frac{1}{h_o A_o}$	11.5	$\text{NTU} = \frac{1}{C_r - 1}\ln\left(\frac{\varepsilon - 1}{\varepsilon C_r - 1}\right) \quad (C_r < 1)$	11.29b

Calculations

Heat Exchanger Project

Professor Nathanael Werner

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Part 1 - Initial Calculations (Tho, Qmax, e, NTU, Area min, Area max, UA ideal)

Given:

	$\dot{m}$ (kg/s)	$T_{m,i}$ (°C)	$T_{m,o}$ (°C)
Hot water	28	90	–
Cold water	27	34	60

Constraints:

- 1m<L<10m
- 850<U<1700

Find:

- Outlet hot water temperature
- UA
- Length of the tube(s)
- Parallel flow or counter flow
- Number of tubes N
- Diameters of tubes
- Cross sectional and surface areas of the heat exchanger
- heat rate of the entire exchanger q
- the effectiveness e
- NTU
- Pressure loss per pipe deltap
- Pump power for flow in all N pipes

Assumptions:

- Steady State
- Incompressible fluid
- Flow is fully developed
- One-dimensional flow
- Constant properties of each fluid (change in properties is negligable)
- Constant pressure specific heat
- Flow is turbulent- Dittus Bolter Nusselt Number
- Mass flow rates remain constant throughout the tubes
- Insulated shell

- Same pressure drop across all pipes
- Tubes are type L
- No fins
- Fouling neglected
- Conduction between inner tubes accounted for
- Radiation neglected
- Overall heat transfer coefficient is constant
- Friction factor is constant
- No changes in kinetic or potential energies
- friction of shell is negligable

```
% Using effectiveness NTU method, the tables, and equations, find
% Temperature of the cold water on average,Cp of the water at that average,
% q (heat transfer rate of the cold side),guess a value for T hot out (not
% given), interpolate with this found t hot out guess to find cphot, use
% that cphot to find Thotout actual and iterate until the actual t hot out
% value is found. from there, determine the properties of the working fluid,
% which is water at the outlet. these properties include mu, k, and the prandtle
% number. then, the Cc and Ch can be determined. Using these values, Cr can be found
% and so can effectiveness. using table 11.2, an ideal UA value can be determined.
% A counter flow vs. parallel flow heat exchanger requires different equations, so
% we must determine the area that suits best.
```

```
%Given
mfh=28; %(kg/s)
mfc=27; %kg/s
tmih=363; %K
tmic=307; %K
tmoc=333; %K
%need to solve for tmoh
surfacerough=0.00003; %m, cited in paper

tcavg= (tmic+tmoc) / 2;
%using this tcavg, look at table a-6
    cpc=4180; %j/kg*k

%material properties cold
muc=interp1([325 330 335], [.000528 .000489 .000453], tmoc, "linear", "extrap");
kc=interp1([325 330 335], [.645 .650 .656], tmoc, "linear", "extrap");
prc=interp1([325 330 335], [3.42 3.15 2.88], tmoc, "linear", "extrap");

%copper properties
rocu=8900; %kg/m^3
kcu=398; %w/m*k
```

$$q = \dot{m}_c c_{p,c}(T_{c,o} - T_{c,i})$$ 11.9

```
%heat rate cold side
qc=mfc * cpc * (tmoc-tmic); %W

% Initial guess for Tmoh
Tmoh_guess1 = 350; % K

% Interpolate for cph from the table
for n = 1:1000  % Max 1000 iterations to avoid infinite loops
    tmhavg = (Tmoh_guess1 + tmih) / 2;
    cph = interp1([275 280 285 290 295 300 305 310 315 320 325 330 335 340 345 350 355 ...
        360 365 370 375 380 385 390 400], ...
        [4211 4198 4189 4184 4181 4179 4178 4178 4179 4180 4182 4184 4186 4188 4191 4195 ...
        4199 4203 4209 4214 4220 4226 4232 4239 4256], ...
        tmhavg, 'linear', 'extrap');
    % J/kgK
```

$$q = \dot{m}_h c_{p,h}(T_{h,i} - T_{h,o})$$ eq 11.6

```
    % Calculate the new temperature of the hot water
    tmoh = tmih - qc / (cph * mfh);

    % Check if the difference between the guess and the calculated tmoh is sufficiently small
    if abs(Tmoh_guess1 - tmoh) <= 0.00001
        break;  % Exit loop if the difference is small enough
    end

    % Update the guess for the next iteration
    Tmoh_guess1 = tmoh;
end

fprintf("Temperature of the hot water at the outlet is %.2f K.\n", tmoh)
```

```
Temperature of the hot water at the outlet is 338.02 K.
```

```
%use this temperature outlet to find fluid properties, table A-6
muh=interp1([330 335 340 345], [.000489 .000453 .000420 .000389], tmoh, "linear", "extrap");
```

```
kh=interp1([330 335 340 345],[.650 .656 .660 .664], tmoh, "linear","extrap");
prh=interp1([330 335 340 345],[3.15 2.88 2.66 2.45], tmoh, "linear","extrap");
rowaterh=980.77;
```

$$\frac{C_{\min}}{C_{\max}} = \frac{\dot{m}_h c_{p,h}}{\dot{m}_c c_{p,c}} = \frac{T_{c,o} - T_{c,i}}{T_{h,i} - T_{h,o}}$$ 11.26

C_r is the *heat capacity ratio* $C_r \equiv C_{\min}/C_{\max}$. page 510

$$q_{\max} = C_{\min}(T_{h,i} - T_{c,i})$$ 11.18

```
%determine heat capacity rates
Cc=mfc*cpc;
Ch=mfh*cph;
Cmin=Cc;
Cmax=Ch;

%heat capacity ratio
Cr=Cmin/Cmax;

%find max possible heat tranasfer rate
qmax=Cmin*(tmih-tmic); %W
fprintf("The maximum heat rate is %.2f. W/m^2 \n", qmax)
```

```
The maximum heat rate is 6320160.00. W/m^2
```

$$\varepsilon \equiv \frac{q}{q_{\max}}$$ 11.19

```
%determine effectiveness
e=qc/qmax;
fprintf("The effectiveness of the heat exchanger is %.2f. \n", e)
```

```
The effectiveness of the heat exchanger is 0.46.
```

$$\mathrm{NTU} \equiv \frac{UA}{C_{\min}}$$ eq 11.24

```
%find Ideal UA for parallel and counter flow
%NTU=UA/Cmin
%table 11.2 page 496
Umin=850; %W/m^2K
Umax=1700; %W/m^2K
```

Flow Arrangement	Relation
Parallel flow	$\mathrm{NTU} = -\frac{\ln[1-\varepsilon(1+C_r)]}{1+C_r}$
Counterflow	$\mathrm{NTU} = \frac{1}{C_r-1}\ln\left(\frac{\varepsilon-1}{\varepsilon C_r-1}\right) \quad (C_r<1)$ $\mathrm{NTU} = \frac{\varepsilon}{1-\varepsilon} \quad (C_r=1)$

Table 11.42

$$\mathrm{NTU} \equiv \frac{UA}{C_{\min}}$$ 11.24

```
%COUNTER FLOW
NTUcf= (1/(Cr-1)) * log((e-1)/(e*Cr-1));
UAidealcf=Cmin*NTUcf;

fprintf("The number of transfer units for counter flow is %.2d.\n", NTUcf)
```

```
The number of transfer units for counter flow is 8.52e-01.
```

```
%determine area of pipes (cf)
Amincf=NTUcf*Cmin/Umax;
Amaxcf=NTUcf*Cmin/Umin;
fprintf("The minimum area of the inside piping is %.2f. m^2 \n", Amincf)
```

```
The minimum area of the inside piping is 56.58. m^2
```

```
fprintf("The maxmimum area of the inside piping is %.2f. m^2 \n", Amaxcf)
```

```
The maxmimum area of the inside piping is 113.16. m^2
```

```
%PARALLEL FLOW
NTUpf= -log(1-(e*(1+Cr))) / (1 + Cr);
UAidealpf=Cmin*NTUpf;

%Counter flow has a much smaller required area to get the same heat
%transfer. Use Counter flow over parallel flow

%determine area of pipes (pf)
Aminpf=NTUpf*Cmin/Umax; %m^2
Amaxpf=NTUpf*Cmin/Umin; %m^2
```

Part 2 - Optimize and Solve (Outstanding variables)

$$Re_D = \frac{4\dot{m}}{\pi D \mu}$$ 8.6

$$Nu_D = 0.023\, Re_D^{4/5} Pr^n$$

where $n = 0.4$ for heating ($T_s > T_m$) and 0.3 for cooling ($T_s < T_m$). These equations have been confirmed experimentally for the range of conditions 8.60

$$D_h \equiv \frac{4A_c}{P}$$ 8.66

$$\frac{1}{UA} = \frac{1}{U_c A_c} = \frac{1}{U_h A_h} = \frac{1}{(hA)_c} + R_w + \frac{1}{(hA)_h}$$ 11.5

$$\frac{1}{\sqrt{f}} = -2.0 \log\left[\frac{e/D}{3.7} + \frac{2.51}{Re_D\sqrt{f}}\right]$$ 8.20

$$\Delta p = -\int_{p_1}^{p_2} dp = f\frac{\rho u_m^2}{2D}\int_{x_1}^{x_2} dx = f\frac{\rho u_m^2}{2D}(x_2 - x_1)$$ 8.22a

$$P = (\Delta p)\dot{\forall}$$ 8.22b

```
n = 38; %number of tubes
```

```
%using L copper pipes (cheaper to use nominal sizes)
D = 0.05; %m
Di2=0.05398; %m
L = 9 ; %m

%Given
mfh=28; %(kg/s)
mfc=27; %kg/s
tmih=363; %K
tmic=307; %K
tmoc=333; %K

tcavg= (tmic+tmoc) / 2; %K
%using this tcavg, look at table a-6
    cpc=4180; %j/kg*K

%material properties cold
muc=interp1([325 330 335], [.000528 .000489 .000453], tmoc, "linear", "extrap"); %N*s/m^2
kc=interp1([325 330 335], [.645 .650 .656], tmoc, "linear", "extrap"); %W/m*K
prc=interp1([325 330 335], [3.42 3.15 2.88], tmoc, "linear", "extrap");
rowaterc=978.2; %kg.m^3

%hot-- use this temperature outlet to find fluid properties, table A-6
muh=interp1([330 335 340 345], [.000489 .000453 .000420 .000389], tmoh, "linear", "extrap"); %
kh=interp1([330 335 340 345],[.650 .656 .660 .664], tmoh, "linear","extrap"); %W/m*K
prh=interp1([330 335 340 345],[3.15 2.88 2.66 2.45], tmoh, "linear","extrap");
rowaterh=980.77; %kg.m^3

%copper properties
rocu=8900; %kg/m^3
kcu=398; %w/m*k

for iN = 1:numel(n)
    Do = 24 .* D; %m Shell diameter
    Aci = (pi*D.^2) / 4 ;%m^2 Cross sectional area of small tubes
    Aco = (pi*Do.^2) / 4 ;%m^2 cross sectional area of shell
    Asi = n*pi*Di2*L; %m^2 Surface area inside tubes
    Aso = pi*Do*L; %m^2 Surface area Annulus, won't need this bc its insulated
    D_2 = 1.05*Di2;   % this is an imaginary diameter around the inner pipes,
                      % factor in the gap between the pipes
    P = pi * (D_2); %m one pipe and its imaginary space
    Ac_2 = n*pi/4*(D_2^2); %m^2 cross-sec area of imaginary gap, total cross sec
                           % gap between all inner tubes

    %velocity through pipe
    Vi = mfh ./ (n*Aci*rowaterh); %m/s

    %reynolds thru pipe
    Rei = 4*mfh ./ (pi *D *muh);
    Reo=4*mfc ./ (pi*Do*muc);
```

```
    %solve for h value inside
    Nui = 0.023 .* Rei .^(4/5) .* prh ^ (0.3);
    hi = Nui .* kh ./ D; %W/m^2*K

    %Hydraulic diameter
    D_hyd = 4*Ac_2/(P*n) ;%m

    %solve for h value outside
    Nuo = 0.023 .* Reo .^(4/5) .* prc ^ (0.4);
    ho = Nuo .* kc ./ D_hyd; %W/m^2*K

    %Conduction resistance - take into account
    Rcond= (log(Di2/D)/ (2*pi*kcu*L)); %ohm

    % calculate U based on both h's and Conduction resistance
    Rtot = (1 ./ (hi*Asi)) + (1 ./ (ho*Aso)) + (Rcond); %ohm
    U=(1/ (Rtot*Asi)); %W/m^2*K

    %Ideal UA
    UAidealcf=UAidealcf; %W/K
    % calculate A
    A=UAidealcf ./ U; %m^2 total surface area


    %Roughness
    epsilonoverD= surfacerough/ D;
    %Friction factor/ use moody or equation
    ff=(1/(-1.8*log10((epsilonoverD/3.7)^1.11 +(6.9/Rei))))^2;

    %Pressure drop
    dp=(ff*rowaterh*Vi^2 / (2*D)) * L ;%N/m^2, Pa
    dptot=dp.*n; %Pa
    totalpumppower= dptot* (mfc/rowaterc); %W
    Pumppowerpipe=totalpumppower/n;


end

fprintf("The number of inner pipes is %.2f .\n", n)
```

```
The number of inner pipes is 38.00 .
```

```
fprintf("The length of pipes is %.2f m.\n", L)
```

```
The length of pipes is 9.00 m.
```

```
fprintf("The diameter of inner pipes is %.2f m.\n", D)
```

```
The diameter of inner pipes is 0.05 m.
```

```
fprintf("The shell diameter is %.2f m.\n", Do)
```

```
The shell diameter is 1.20 m.
```

```
fprintf("The cross sectional area of inner pipes is %.2d m^2.\n", Aci)
```

```
The cross sectional area of inner pipes is 1.96e-03 m^2.
```

```
fprintf("The cross sectional area of the shell is %.2d m^2.\n", Aco)
```

```
The cross sectional area of the shell is 1.13e+00 m^2.
```

```
fprintf("The total surface area of inner pipes is %.2f m^2.\n", Asi)
```

```
The total surface area of inner pipes is 58.00 m^2.
```

```
fprintf("The total surface area of shell is %.2f m^2.\n", Aso)
```

```
The total surface area of shell is 33.93 m^2.
```

```
fprintf("The velocity in one inner tube is %.2f m/s.\n", Vi)
```

```
The velocity in one inner tube is 0.38 m/s.
```

```
fprintf("The reynolds number of inner pipes is %.2f .\n", Rei)
```

```
The reynolds number of inner pipes is 1646441.69 .
```

```
fprintf("The reynolds number of fluid in shell is %.2f .\n", Reo)
```

```
The reynolds number of fluid in shell is 61292.02 .
```

```
fprintf("The ideal UA value is %.2f W/K .\n", UAidealcf)
```

```
The ideal UA value is 96184.94 W/K .
```

```
fprintf("The total surface area of the heat exchanger is %.2f m^2.\n", A)
```

```
The total surface area of the heat exchanger is 80.67 m^2.
```

```
fprintf("The friction factor is %.2f .\n", ff)
```

```
The friction factor is 0.02 .
```

```
%minimizing pump power decreases cost significantly over the years. Might
%be more expensive up front with more area, but running it and longevity
%top that.
fprintf("The change in pressure in each pipe is %.2f Pa.\n", dp)
```

```
The change in pressure in each pipe is 228.43 Pa.
```

```
fprintf("The change in pressure overall is %.2f Pa.\n", dptot)
```

```
The change in pressure overall is 8680.47 Pa.
```

```
fprintf("The total pump power is %.2f W.\n", totalpumppower)
```

```
The total pump power is 239.60 W.
```

```
fprintf("The pump power per pipe is %.2f W.\n", Pumppowerpipe)
```

```
The pump power per pipe is 6.31 W.
```

```

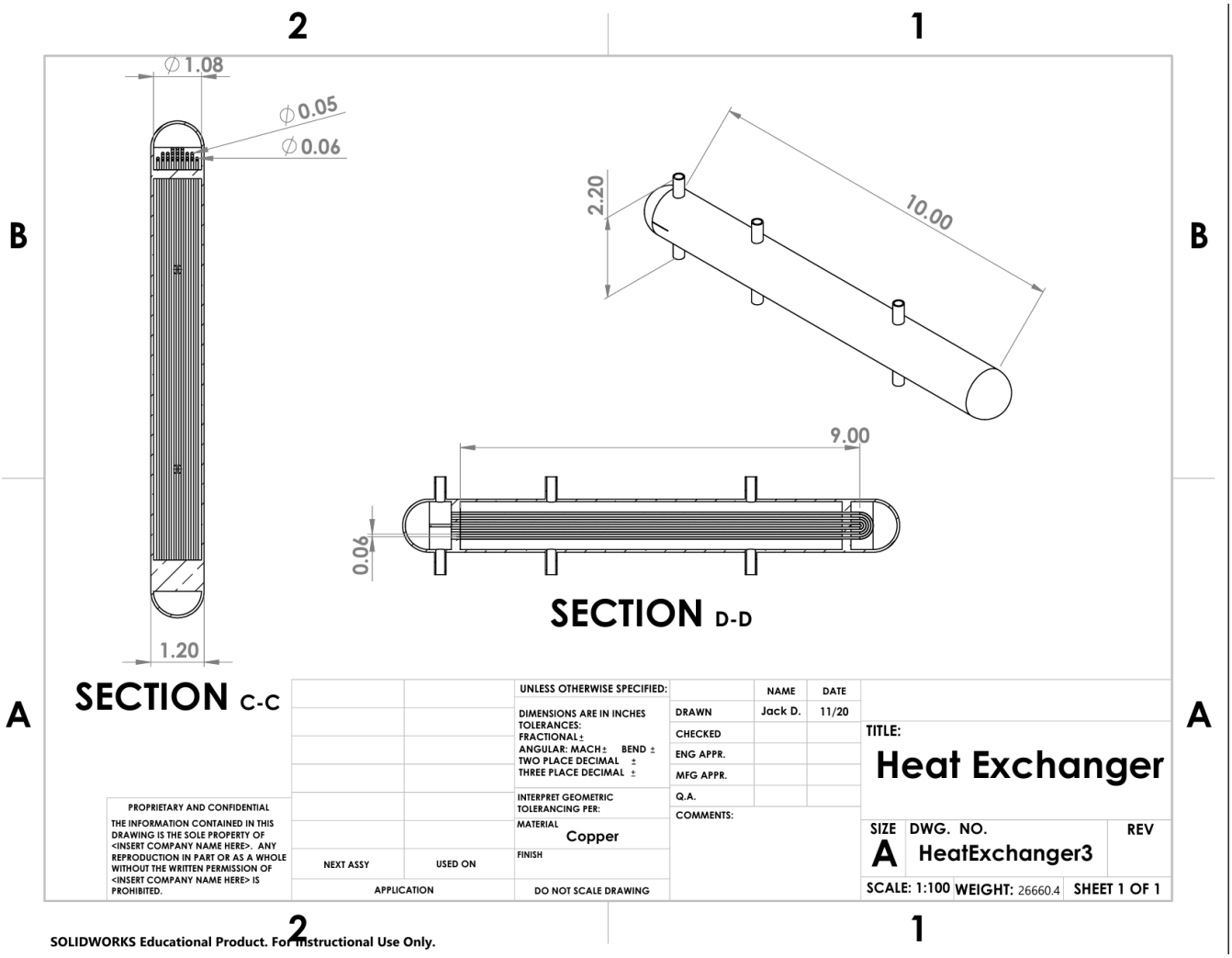
```

Part 3 - Cost Analysis

```
%Counter flow is more efficient and less expensive
%Using copper tyle L will help cut down expenses from a customized pipe
%Pricing metric cited in paper
Priceinnertube=n* L* 4.699 %$
```

```
Priceinnertube =
     1.607058000000000e+03
```

```
%Will be at least this, plus the custom outer shell cost.
```

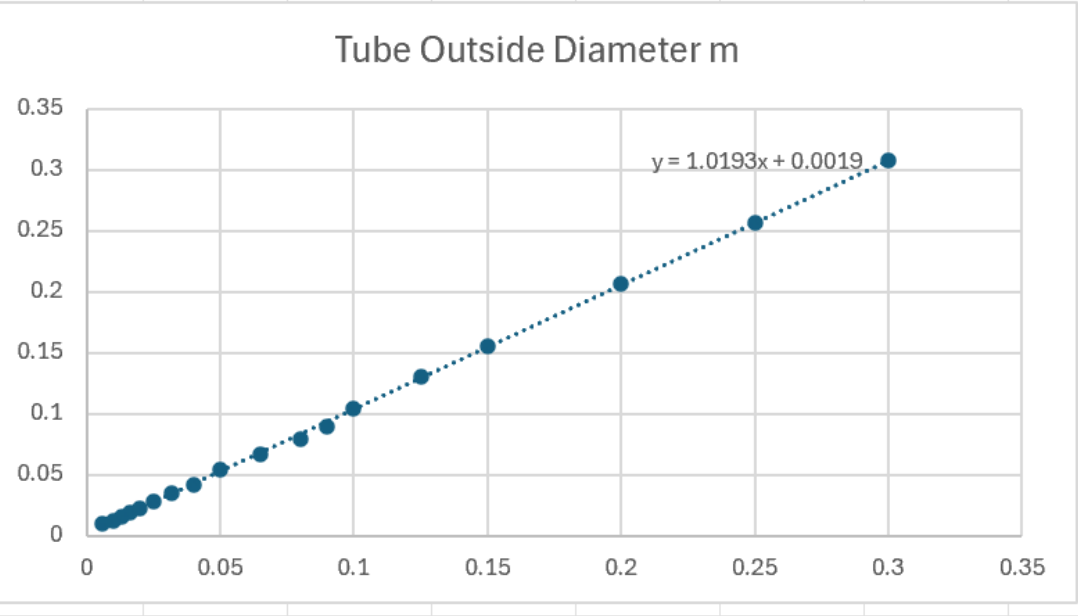

2
1
B
A
∅1.08
∅0.05
∅0.06
1.20
SECTION C-C
2.20
10.00
9.00
0.06
SECTION D-D
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TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
Copper
FINISH
DO NOT SCALE DRAWING
NAME
DATE
DRAWN
Jack D.
11/20
CHECKED
ENG APPR.
MFG APPR.
Q.A.
COMMENTS:
TITLE:
Heat Exchanger
SIZE
A
DWG. NO.
HeatExchanger3
REV
SCALE: 1:100
WEIGHT: 26660.4
SHEET 1 OF 1
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APPENDIX D

Excel sheets calculations and tables

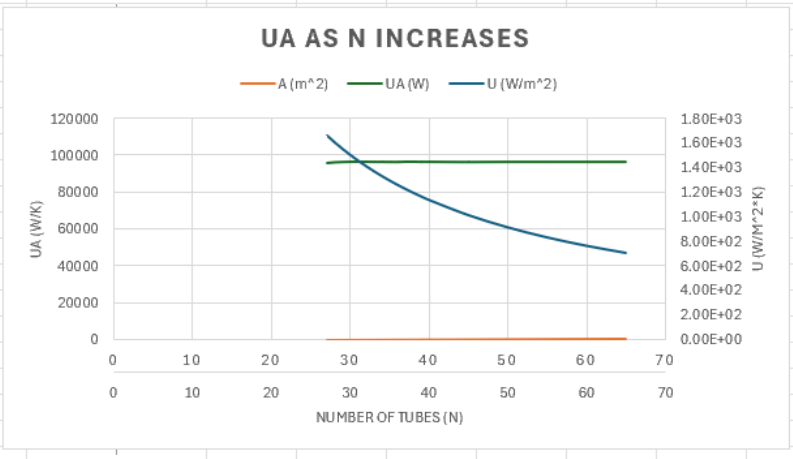
Nominal Tube Size Inches	Nominal Tube Size m	Tube Outside Diameter Inches	Tube Outside Diameter m	Type	Wall Thickness Inches	Weight of Tube Lb/ft	Weight of Tube & Water Lb/ft
1/4"	0.006	0.375	0.00953	L	0.03	0.126	0.16
3/8"	0.01	0.5	0.0127	L	0.035	0.198	0.26
1/2"	0.013	0.625	0.01588	L	0.04	0.285	0.385
5/8"	0.016	0.75	0.01905	L	0.042	0.362	0.513
3/4"	0.02	0.875	0.02223	L	0.045	0.455	0.664
1"	0.025	1.125	0.02858	L	0.05	0.655	1.012
1-1/4"	0.032	1.375	0.03493	L	0.055	0.884	1.43
1-1/2"	0.04	1.625	0.04128	L	0.06	1.14	1.91
2"	0.05	2.125	0.05398	L	0.07	1.75	3.091
2-1/2"	0.065	2.625	0.06668	L	0.08	2.48	4.544
3"	0.08	3.125	0.07938	L	0.09	3.33	6.279
3-1/2"	0.09	3.625	0.09008	L	0.1	4.29	8.279
4"	0.1	4.125	0.10478	L	0.11	5.38	10.568
5"	0.125	5.125	0.13018	L	0.125	7.61	15.691
6"	0.15	6.125	0.15558	L	0.14	10.2	21.816
8"	0.2	8.125	0.20638	L	0.2	19.28	39.579
10"	0.25	10.125	0.25718	L	0.25	31.1	61.69
12"	0.3	12.125	0.30798	L	0.28	40.4	85.826

https://taylorwalraven.ca/copper-tubing-data-Type-L.php
Cited in Paper



Temperature of the hot water outlet (K, c)	*338.02, 65.02*	*358.73, 85.73*
Maximum heat rate (W/m^2)	*6320160*	*33783.7*
Effectiveness	*0.4643*	*0.584481*
NTU	*0.852*	*0.940509*
Minimum Area of inside tubing (m^2)	*56.58*	*mfhi=.7369*
Maximum Area of inside tubing (m^2)	*113.16*	*mfci=0.14394*
Number of inner pipes	*38*	
Length of pipes (m)	*9*	
Outer diameter of inner pipes (m)	*0.05398*	
Inner diameter of inner pipes (m)	*0.05*	
Shell Diameter (m)	*1.2*	
cross sectional area of inner pipes (m^2)	*1.96E-03*	
Cross sectional area of Shell (m^2)	*1.13E+00*	
Total surface area of inner pipes (m^2)	*58*	
Total surface area of outer pipes (m^2)	*33.93*	
Velocity in one inner tube (m/s)	*0.38*	
Reynolds inner pipe	*1646441.69*	
Reynolds outer pipe	*61292.02*	
Ideal UA value (W/K)	*96184.94*	
Total surface area of heat exchanger (m^2)	*80.67*	
Friction factor	*0.02*	
Change in pressure in each pipe (Pa)	*228.43*	
Change in pressure overall (Pa)	*8680.47*	
Total pump power (W)	*239.6*	
Pump power per pipe (W)	*6.31*	
MATLAB	ANSYS	ANSYS
model	model	validation
	One tube	One tube

N	U (W/m^2)	A (m^2)	UA (W)		Vary from U=850 to U=1700	56.58<A<113.56
27	1.66E+03	58.0309	9.62E+04	Goal:	Reducing pump power required	
28	1.60E+03	60.0902	9.62E+04			
29	1.55E+03	62.1534	9.62E+04			
30	1.50E+03	64.2107	9.62E+04			
31	1.45E+03	66.2679	9.62E+04			
32	1.41E+03	68.3251	9.62E+04			
33	1.37E+03	70.3823	9.62E+04			
34	1.33E+03	72.4395	9.62E+04			
35	1.29E+03	74.4967	9.62E+04			
36	1.26E+03	76.5539	9.62E+04			
37	1.22E+03	78.6111	9.62E+04			
38	1.19E+03	80.6683	9.62E+04			
39	1.16E+03	82.7256	9.62E+04			
40	1.13E+03	84.7828	9.62E+04			
45	1.01E+03	95.0688	9.62E+04			
50	9.13E+02	105.3549	9.62E+04			
55	8.32E+02	115.6409	9.62E+04			
60	7.64E+02	125.927	9.62E+04			
65	7.06E+02	136.2131	9.62E+04			



Diamete	U	A	UA
0.006	4.16E+03	23.1209	9.62E+04
0.01	5.17E+03	18.5964	9.62E+04
0.013	4.47E+03	21.5186	9.62E+04
0.016	3.87E+03	24.8414	9.62E+04
0.02	4.12E+03	23.327	9.62E+04
0.025	2.58E+03	37.3389	9.62E+04
0.032	2.28E+03	42.2673	9.62E+04
0.04	2.19E+03	43.8344	9.62E+04
0.05	1.19E+03	80.6683	9.62E+04
0.065	9.86E+02	97.5139	9.62E+04
0.08	7.09E+02	135.5859	9.62E+04
0.09	6.24E+02	154.1027	9.62E+04
0.1	4.27E+02	225.2992	9.62E+04
0.125	2.97E+02	323.8555	9.62E+04
0.15	2.19E+02	438.8129	9.62E+04
0.2	1.34E+02	715.7052	9.62E+04

Area increases as more tubes are added (obviously), but this decreases the overall heat transfer coefficient

